Partial Multiview Incomplete Multilabel Learning via Uncertainty-Driven Reliable Dynamic Fusion

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***Abstract*—Currently, an increasing number of researchers are focusing on partial multiview incomplete multilabel learning. However, many methods generally integrate features from multiple views via an average weighting strategy, which overlooks the potential mismatch between the contribution of each view and their assigned fusion weights and thus generates unreliable fused features. To address this issue, we propose a novel uncertainty-driven reliable dynamic fusion framework for partial multiview incomplete multilabel learning. Unlike existing methods, the proposed uncertainty-driven reliable sample-level dynamic fusion module operates on the principle that samples exhibiting greater uncertainty possess fewer reliable features. This module evaluates the uncertainty of each sample and, in turn, estimates the reliability of features with the uncertainty of sample judgement, thereby obtaining reliable weights to guide the information fusion of multiple views. Furthermore, many existing approaches for handling incomplete multilabel scenarios typically concentrate on the information from annotated labels, neglecting the potential information of unknown tags. To bridge this gap, we incorporate an innovative pseudolabelling strategy that effectively identifies trustworthy pseudolabels that correspond to those unannotated uncertain labels, thereby adding additional supervisory information to assist model training. Moreover, we also devise a feature masking strategy to further augment the encoder's representation learning capabilities. The experimental results across five datasets demonstrate that our method outperforms current state-of-the-art methods.**

***Index Terms*—Partial multiview learning, incomplete multilabel learning, multilabel classification, multiview uncertainty.**

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I. Introduction

In THE rapidly evolving field of machine learning, multiview multilabel learning has emerged as a compelling subject. This field is adept at handling complex and rich real-world data, which often present itself in multiple heterogeneous forms or 'views', each view is a different description of the research objective [1], [2], [3], [4], [5]. The introduction of multiview multilabel learning (MvMlL) has far-reaching implications, from boosting the performance of recommendation systems by leveraging user-item interactions across various domains to advancing biomedical research through the integration of multiomics data for disease diagnosis and treatment [6], [7], [8]. As the volume and variety of data continue to grow, the need for robust, reliable, and interpretable multiview multilabel learning models becomes more pronounced. For example, Liu et al. developed a matrix completion-based multiview learning model for image classification, termed low-rank multiview matrix completion (lrMMC). The lrMMC model adeptly seeks a shared, low-dimensional representation by harmonizing the diverse information presented across different views [9]. Afterwards, Zhang et al. proposed an approach that leverages matrix factorization to capitalize on the synergistic potential of different views [10]. Recently, Sun and colleagues proposed the latent conditional Bernoulli mixture (LCBM) model, a sophisticated multiview probabilistic model for multilabel classification. The LCBM model stands out as a generative approach that integrates inputs from various views; it ingeniously employs a shared latent space to underpin a Bernoulli mixture model, which captures and models the interdependencies among labels [11].

In general, the above conventional methods assume that the data available for model design and training are complete in terms of views and labels. However, in practice, multiview multilabel learning faces a significant challenge: owing to the complexity of data collection and label annotation, the resulting dataset is often both partially multiview and incompletely labelled [12], [13], [14], [15]. This implies that not all views are available for every instance, and some labels may be missing or incorrectly annotated, thereby solving the problem of partial multiview incomplete multilabel learning (PMvIMlL). Consequently, traditional multiview multilabel learning methods may struggle to perform well under these conditions. To address this challenge, researchers have proposed various strategies. Xu et al. introduced a groundbreaking method termed MVL-IV, which tackles the complex task of multiview learning under incomplete

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view data conditions [16]. The method is predicated on the assumption that the various views, despite being potentially incomplete, are generated from a common underlying subspace. Zhang et al. introduced a model called CPM-Nets, which is designed to learn a latent multiview representation by emulating the process of data transmission [17]. This innovative approach aims to strike an optimal balance between the consistency and complementarity of information across different views. However, the aforementioned methods tend to address only partial multiview challenges. Similarly, there are approaches that concentrate on addressing incomplete multilabel issues while neglecting certain multiview scenarios. For example, Yeh et al. proposed a model termed C2AE, which is founded on a deep neural network, to explore and construct a potential space [18]. A key feature of C2AE is its innovative approach to joint features and tag embedding. After that, Ma et al. crafted a straightforward yet potent approach known as discriminant multilabel learning (DM2L) that cleverly imposes a low-rank structure on the predictions made for instances that share identical tags and employs a high-rank structure to ensure that the diversity among predictions is preserved [19].

This above selective focus may limit the comprehensiveness of the solutions offered. Therefore, Tan et al. proposed iMVWL to address the learning issue under both partial multiview and incomplete multilabel cases [20]. The iMVWL model adeptly projects partial multiview features into a unified potential space via the matrix decomposition mechanism. Furthermore, the model leverages a subsequent round of matrix decomposition to establish a clear and direct connection between the feature space and the label space. However, the above traditional machine learning models have inherent feature extraction limitations and often struggle to predict labels for new samples. To address these limitations, Wen et al. introduced a deep learning-based model named DD-IMvMLC-net [21]. This model is designed to capture the high-order semantic features of the samples, enabling an end-to-end multilabel classification approach that enhances the predictive power and generalization capabilities of the model. Liu et al. subsequently advanced the field by proposing two innovative models, DIC-net and MTD [22], [23]. DIC-net extends the capabilities of DD-IMvMLC-net with a comparative learning mechanism that focuses on reducing the distance between the same sample's representations in various views and expanding it between different sample pairs, thereby enhancing the discriminability of features. On the other hand, MTD introduces a novel approach to feature extraction by differentiating between the shared and private features of views; this allows MTD to capitalize on the shared and unique information presented in the data, leading to improved classification accuracy and a deeper understanding of sample characteristics.

However, a limitation exists across the aforementioned methods: they uniformly employ an average weighting strategy to integrate features from multiple views, which results in the problem that the contribution of each view does not match the corresponding fusion weight. Therefore, we propose an uncertainty-driven reliable dynamic fusion model to address the PMvIMlL challenge, which is abbreviated as URDF. Specifically, we propose an uncertainty-driven reliable sample-level dynamic fusion strategy that addresses the problem of incongruity between the contributions of individual views and their fusion weights. More importantly, this strategy provides detailed processing for each sample instead of learning the overall weight of each view. The rationale behind this fusion strategy is inspired by the principle that as the uncertainty in a sample's classification increases, the reliability of sample features decreases. Furthermore, considering that the distribution of labels in multilabel learning often exhibits a long-tail distribution since some samples generally have a small number of positive labels and a large number of negative labels [24], [25], [26], [27], we introduce an innovative pseudolabelling strategy to address the insufficient label information issue caused by missing labels. The proposed method can select credible negative labels to compensate for the missing labels, thereby training the model with extra reliable supervisory information. In addition, drawing inspiration from successful masking techniques [23], [28], [29], [30], we introduce a new masking strategy to enhance the encoder's learning capabilities. Finally, our model is adept at handling not only partial multiview incomplete multilabel learning scenarios but also complete multiview multilabel learning tasks.

In summary, our contributions are distilled into the following key points:

- To address the challenge of the partial multiview incomplete multilabel learning task, we present a novel uncertainty-driven reliable sample-level dynamic fusion strategy that links feature reliability with sample judgement uncertainty to solve the problem of a mismatch between the contributions of individual views and their fusion weights. This strategy also meticulously learns the weight of each sample within all views instead of learning the overall weight of each view.
- We propose a pseudolabelling approach that effectively selects trustworthy pseudolabels for the unknown of training sample labels, thereby enriching the supervisory information for model training.
- We also introduce a novel masking strategy, which is designed to enhance the encoder's representation learning ability. Our method has been tested on five widely used datasets and has shown superior experimental performance compared with other methods.

II. Related Work

In this section, we introduce the relevant works on PMvIMlL.

A. MvMlL

Multiview multilabel learning (MvMlL) addresses the challenge of classifying instances based on multiple descriptive features (views), where each instance may be associated with multiple labels. For example, the lrMMC model employs matrix completion techniques for multiview multilabel image classification [9]. It commences by learning a low-rank common representation from the multiview features, subsequently utilizing the matrix completion approach to deduce the labels for each sample. Zhao et al. contributed to this field with the introduction

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of the CDMM and LSVL models. The CDMM model considers the consistency, diversity, and label correlation inherent in multiview learning, employing an ensemble learning strategy to predict final model outcomes [31]. On the other hand, the LSVL model operates on the premise of structural consistency between the view and label spaces [32]; it sequentially constructs view-specific tag models and delves into the consistency information from multiple perspectives. Furthermore, LSVL uncovers the low-rank correlation information embedded among multiple tags, thereby enriching the information contained in the original tag space.

B. PMvL or IMlC

Partial multiview learning (PMvL) and incomplete multilabel classification (IMlC) address only one of the two missing cases.

In the realm of PMvL, Xue et al. introduced the DCPSL method, which unifies semisupervised depth matrix decomposition, correlation subspace learning, and multiview label prediction to jointly learn shared and private label predictors within a depth potential subspace [33]. Zhang et al. proposed CPM-net, which transforms the task of learning potential multiview representations into a degeneration process by simulating data transmission and achieves an optimal equilibrium between the consistency and complementarity of information across different views [17]. Building upon CPM-net, Zhang et al. further advanced the model's capabilities [34]. By employing an adversarial strategy, the model can stably fill the missing perspectives and encode all the perspective information of each sample into the potential representation to further enhance its integrity.

In the context of IMlC, C2AE employs canonical correlation analysis and an automatic encoder to achieve end-to-end learning via the effective utilization of tag dependencies [18]. In addition, Zhu et al. developed GLOCAL, which is capable of learning label representations and optimizing the label manifold while leveraging global and local label correlations to address full and missing label scenarios [35]. Following this, the GLOCAL model was extended to a multiview version, resulting in the GLMVML model [36].

C. PMvIMlL

As the term implies, partial multiview incomplete multilabel learning includes the completion of two tasks at the same time: partial multiview learning and incomplete multilabel classification. The field has evolved through three primary technological approaches: matrix decomposition, multilayer perceptron (MLP), and transformer architectures.

Tan et al. proposed iMVWL, which considers two types of missing cases at the same time for the first time [20]. iMVWL maps multiview data to the label space through two-stage matrix decomposition and eliminates the negative effects of missing views and missing labels with the help of a missing indication matrix. This method cleverly uses a missing indication matrix to counteract the effects of missing views and labels. Subsequent work by Li et al. introduced NAIML, which considers the high-rank structure of global labels and the low-rank structure of local labels to enhance predictive classification [37].

Neural network integration into PMvIMlL was spearheaded through the DD-IMvMLC-net model based on the MLP model proposed by Wen et al. [21], and Liu et al. subsequently introduced comparative learning into DD-IMvMLC-net. This method can help the model capture more consistent information by reducing the distance between the same sample features and increasing the distance between different samples [22]. Liu et al. subsequently enriched this domain with models such as MTD and AIM-net; MTD focuses on the extraction of shared and private view information to increase the ability of the model to capture more semantic information [23]. AIM-net aims to complete incomplete multiview data by approximating missing features through cross-view joint attention [38]. Long et al. proposed a method called MSLPP to keep the semantic structure information between samples during feature extraction [39]. Liu et al. proposed the RANK model to learn reliable multiview feature representations [40]. Furthermore, Liu et al. proposed LMVCAT based on the transformer. LMVCAT focuses on the information interaction between multiple views and restricts feature learning in multiple views through the relationships of samples in the label space [41]. Subsequently, Ou et al. merged a two-layer transformer module to describe the interaction between views and labels and adopted a KNN-style missing view generation module to fill in missing views [42]. In addition, Xie et al. proposed UPDGD-Net, which aims at maintaining the internal of the data in the feature extraction stage while exploring more supervision information about labels [43].

Our approach, which is predicated on multilayer perceptron technology, has distinct advantages over existing methodologies, such as LMVCAT and DD-IMvMLC-net. First, our model architecture is relatively straightforward and intuitive. Second, our model addresses the limitations of current models that fail to provide nuanced guidance for cross-view fusion of each sample within each view. Finally, our model fully exploits the latent value of missing labels, thereby enhancing the overall performance and robustness of the model.

III. Proposed Method

In this section, we initially provide a comprehensive description of the PMvIMlL issue. We subsequently elaborate on the methodology that we propose.

A. Main Notation and Problem Description

We can represent the set of original multiview data as $\{X^{(v)}\}_{v=1}^{m}$ and the associated multilabel as Y. Here, m denotes the total number of views. The feature set for the v-th view of the data is represented by $X^{(v)} \in R^{n\times d_v}$, where n is the number of instances and d_v is the dimension of the feature space for instances within the v-th view. For the i-th particular instance, its feature in the v-th view is denoted as $X_i^{(v)}$. The label space is denoted by the matrix $Y \in \{0,1\}^{n\times c}$, where c is the total label count. In this matrix, if $Y_{i,j} = 1$, the i-th instance has the j-th label, and if $Y_{i,j} = 0$, the i-th instance

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TABLE I
THE SUMMARY OF NOTATIONS IN THIS PAPER

Symbols	Meanings
n	The number of samples
m	The number of views
c	The number of categories
d_v	The dimension of the i-th view
d	The dimension of the latent subspace
$X^{(v)}$	The original features in the v-th view
Y	The true label matrix
W	View missing indication matrix
G	Label missing indication matrix
$M^{(v)}$	The feature mask matrix of the v-th view
l_v	The number of mask features in the v-th view
μ	The masking rate of features
$\overline{X}^{(v)}$	The features in the v-th view after the masking strategy
$E^{(v)}$	The encoder in the v-th view
$\widetilde{X}^{(v)}$	The features extracted by the encoder in the v-th view
Z	The fusion feature
$\overline{Z}$	The ultimate fusion representation
q_v	A trainable scalar weight in the view v
γ	An adjustment factor
$P^{(v)}$	The prediction probability matrix in view
P	The prediction probability matrix of the fusion feature
$Q^{(v)}$	The quality score matrix for the features of the instance in the v-th view
Q	The quality score matrix for the view
$\rho_i^{(v)}$	The quality score of features of the i-th sample in the v-th view
Φ	The classifier
θ	The sigmoid activation function
$H^{(v)}$	Indication matrix of valid labels in the v-th view
$\widehat{X}^{(v)}$	The reconstructive feature of the v-th view
$D^{(v)}$	The decoder in the v-th view
U	The loss matrix
ϱ_i	The average loss of the labels existing in the i-th sample
G^{t-}	The indicator matrix of trustworthy negative labels
U^v	The loss matrix of the v-th view
ϱ_i^v	The average loss of the real label of the i-th sample
G^{v-}	The indication matrix of the selected pseudo label in the v-th view
α	The penalty coefficients

does not have the j-th label. We also introduce another matrix $W \in \{0,1\}^{n\times m}$ to indicate missing views, where $W_{i,v}=1$ means the i-th instance is present in the v-th view, and $W_{i,v}=0$ means the v-th view of the i-th instance is missing. Similarly, the matrix $G \in \{0,1\}^{n\times c}$ is used to track known labels, where $G_{i,j}=1$ means the j-th label's information for the i-th instance is available, and $G_{i,j}=0$ means it is missing. During data preprocessing, missing elements within the feature $\{X^{(v)}\}_{v=1}^m$ and the label matrix Y are assigned a value of 0. The ultimate objective of PMvIMlL is to construct a predictive model that can accurately infer multiple labels for each instance, even in the presence of incomplete multiview information.

To facilitate comprehension and ease of reference, we have compiled a comprehensive summary of the symbols and their corresponding meanings in Table I.

B. The Proposed Method

The architecture of our model is presented in Fig. 1, which is composed of four modules: a feature extraction module with masked fragments, an uncertainty-driven reliable sample-level dynamic fusion module, a feature reconstruction module, and an incomplete multilabel classification module with a pseudolabel strategy.

1) Feature Extraction Module With Masked Fragments: Building on the remarkable success of the masking automatic encoder (MAE) [28] in image analysis and the success of VideoMAE in video processing [29], MTD employs a comparable masking strategy to bolster the encoder's learning capabilities [23]. However, the MTD operation deactivates the features in a region, which leads to the loss of block information. Our goal is to apply a similar masking mechanism to multiview data without losing a large area of information. Consequently, we introduce an innovative masking strategy that targets the inactivation of feature points, not feature blocks, to preserve more granular information. In particular, m matrices $\{M^{(v)} \in \{0,1\}^{n\times d_v}\}_{v=1}^m$ are constructed to signify which sample features are masked in the v-th view. Initially, all values in the matrix $M^{(v)}$ are set to 1. The length of the masked feature in the v-th view is subsequently computed by $l_v = \mu * d_v$, where μ is the rate at which the features are masked. Afterwards, l_v values in the range of $\{0, d_v\}$ are randomly generated for each instance in each view and represented by $\{\{\{A_{i,j}^{(v)}\}_{i=1}^n\}_{j=1}^{l_v}\}_{v=1}^m$, which indicates that the $A_{i,j}^{(v)}$-th feature element for the i-th sample in the v-th view is masked; this procedure can be expressed as follows: $\{\{\{M_{i,A_{i,j}^{(v)}}^{(v)}\}_{i=1}^n\}_{j=1}^{l_v}\}_{v=1}^m = 0$. These matrices store a mask segment that begins at random bits, leading to partial deactivation of the feature vectors. The original input data are masked via m mask matrices, and the corresponding masked feature matrices are obtained as follows:

$$\overline{X}^{(v)} = X^{(v)} \odot M^{(v)} \tag{1}$$

where $\overline{X}^{(v)} \in R^{n\times d_v}$ denotes the masked feature of the v-th view and where $\odot$ indicates elementwise multiplication.

The feature dimensions of all views are generally different due to their diversity, which is not conducive to the parallel execution of the network. To address this issue, we employ the encoder network to map the multiview data into a semantic space with a unified feature dimension d. Similar to [23], we employ a set of MLPs to serve as the encoder for each view. This is expressed as $\{E^{(v)} : \overline{X}^{(v)} \to \widetilde{X}^{(v)}\}_{v=1}^m$, where $E^{(v)}$ represents the encoder of the v-th view and where $\widetilde{X}^{(v)} \in R^{n\times d}$ is the extracted feature with rich semantic information from the v-th view.

2) Uncertainty-Driven Reliable Sample-Level Dynamic Fusion Module: By integrating $\widetilde{X}^{(v)}$, which represents each view, we can perform a cross-fusion of multiple perspectives to obtain a unique representation. The majority of multiview learning approaches rely on either combining or aggregating features from various views to generate a unified representation of the samples [44], [45], [46]. Nonetheless, such techniques may not be appropriate for partial multiview scenarios. Therefore, some researchers have attempted to fuse multiple views with the average weighting strategy as follows [21], [47]:

$$Z_i = \sum_{v=1}^{m} \frac{W_{i,v}\widetilde{x}_i^{(v)}}{\sum_{v'=1}^{m} W_{i,v'}} \tag{2}$$

where $Z \in R^{n\times d}$ is the fusion feature and where $\widetilde{x}_i^{(v)}$ represents the feature vector of the i-th instance in $\widetilde{X}^{(v)}$. However, this method ignores the problem that the information in each view has a different quality level. Other researchers subsequently

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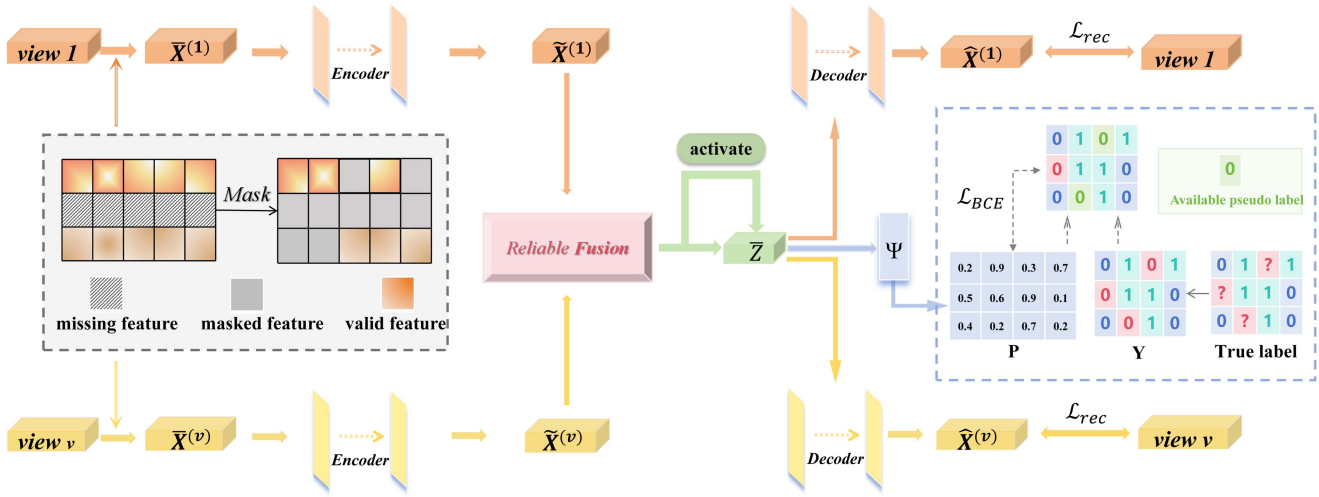


Fig. 1. The architecture diagram of the URDF model. $\overline{Z}$ indicates the fused features. The 'activate' function is a sigmoid activation function. Ψ is the classifier.

proposed the following adaptive fusion method [41], [48]:

$$Z_i = \sum_{v=1}^{m} \frac{e^{q_v^{\gamma}} \widetilde{x}_i^{(v)} W_{i,v}}{\sum_{v'} e^{q_{v'}^{\gamma}} W_{i,v'}} \tag{3}$$

where q_v represents a trainable scalar weight that is specific to the v-th view, indicating the importance or influence of that particular view in the learning process. γ serves as an adjustment factor. However, this approach lacks specific guidance during the process of learning the weights of the views, which may result in unreliable weights. Furthermore, the resulting weights represent the overall importance of a view but fail to provide a detailed treatment for each individual sample. To address this issue, we propose a new uncertainty-driven sample-level dynamic fusion strategy to obtain reliable cross-view fusion features for instances. This strategy is intuitive: as the uncertainty of sample judgement increases, the reliability of sample features decreases. Our approach provides fine-grained guidance for the cross-view fusion of samples. This allows for the appropriate allocation of weights based on the uncertainty of sample features, which is effective during both the training phase and the testing phase.

First, the probability of each sample's category is predicted by the features in each view: $\{P^{(v)} = \Phi(\widetilde{X}^{(v)})\}_{v=1}^{m}$; here, $P^{(v)} \in [0,1]^{n\times c}$ signifies the predictive probabilities of samples derived from the characteristics in the v-th view, and c is the number of labels. Φ represents a classifier. The classifier consists of a fully connected layer with a sigmoid activation function. We subsequently convert the prediction probability of the sample into the quality of the sample features via the following function:

$$Q_{i,j}^{(v)} = 1 - P_{i,j}^{(v)} \left(1 - P_{i,j}^{(v)}\right) \tag{4}$$

where $Q^{(v)} \in [0,1]^{n\times c}$ represents the quality score matrix for the features of the instance in the v-th view. $P_{i,j}^{(v)}\left(1 - P_{i,j}^{(v)}\right)$ measures the uncertainty of the prediction about the j-th label of the i-th sample. Then, we calculate the average quality scores of the features for each instance in the v-th view.

$$\rho_i^{(v)} = \left(\sum_{j=1}^{c} Q_{i,j}^{(v)}\right) / c \tag{5}$$

where $\rho^{(v)} \in [0,1]^{n\times 1}$ and where $\rho_i^{(v)}$ is the quality score of features of the i-th sample in the v-th view. The process of normalizing the uncertainty by dividing by c ensures that the quality score, which reflects the reliability of the sample features, is scaled appropriately. Furthermore, considering the range of uncertainty, which inherently lies between 0 and 1, the quality score of the samples also falls within this interval after normalization. This range is significant because it aligns with the probabilistic nature of uncertainty measurements. In other words, the quality score can indicate the reliability of sample features. Next, we combine the quality scores of all samples across m views. To mitigate the impact of missing views, we also incorporate the view-missing indication matrix W. The effective quality scores of features for samples in all views $Q \in [0,1]^{n\times m}$ can be obtained via the following formula:

$$Q = concat\left(\rho^{(1)}, \ldots, \rho^{(m)}\right) \odot W \tag{6}$$

where $concat()$ represents the connection operation.

We subsequently normalize the quality score of each sample across all views and use it as the weight for the sample in each view:

$$Q_{i,v} = Q_{i,v} / \sum_{v=1}^{m} Q_{i,v} \tag{7}$$

An intuitive example of a learning weight is shown in Fig. 2. After that, we integrate the weight matrix to fuse the features of multiple views.

$$Z_{i,j} = \sum_{v=1}^{m} \widetilde{X}_{i,j}^{(v)} * Q_{i,v} \tag{8}$$

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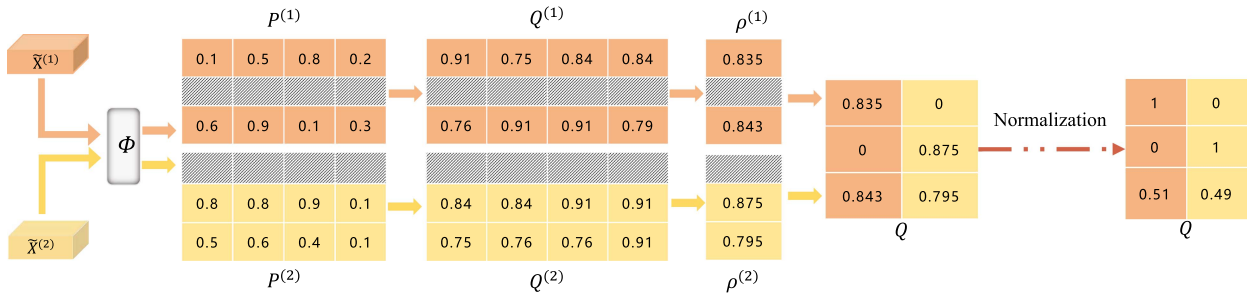


Fig. 2. Schematic diagram for the view weight learning process, taking the fusion of two views as an example.

where $Z \in R^{n\times d}$ is the feature of samples after cross-view fusion.

In addition, on the basis of [23], we introduce a strategy to realize the self-enhancement of the fusion features:

$$\overline{Z} = Z \odot \theta(Z) \tag{9}$$

where θ indicates a sigmoid activation function and where $\overline{Z} \in R^{n\times d}$ denotes the ultimate fusion representation.

Nevertheless, the reliable fusion representation mentioned relies on the dependable sample weights learned from accurate $P^{(v)}$; thus, providing adequate supervision for the classifier Φ to ensure the precision of the prediction $P^{(v)}$ is crucial. To this end, we employ the ground truth to guide and optimize the classifiers via a binary cross-entropy (BCE) loss function, thereby enhancing the reliability of the classification process. Given the presence of partial multiview incomplete multilabel data, the direct application of the BCE function is not feasible. For each individual view, we focus solely on the valid samples and their corresponding effective labels. To formalize this, we introduce the index matrix $H(v)$, which represents the valid label indices for view v:

$$H_{i,j}^{(v)} = G_{i,j} W_{i,v} \tag{10}$$

where $H^{(v)} \in \{0,1\}^{n\times c}$, n is the number of samples and L is the number of labels. Within this matrix, $H_{i,j}^{(v)} = 1$ indicates that the j-th label for the i-th sample in the v-th view is valid. If $H_{i,j}^{(v)} = 0$, the label is not valid, which could be due to the absence of the sample in that view or the unavailability of the label. In other words, $H_{i,j}^{(v)} = 1$ confirms both the presence of the sample in the view v and the presence of the label for that sample without any missing data. Consequently, we adopt the following weighted BCE function to supervise the model training:

$$\mathcal{L}_{BCE_v} = -\frac{1}{mnc}\sum_{v=1}^{m}\sum_{i=1}^{n}\sum_{j=1}^{c}\left(Y_{i,j}\log\left(P_{i,j}^{(v)}\right) + (1-Y_{i,j})\log\left(1-P_{i,j}^{(v)}\right)\right)H_{i,j}^{(v)} \tag{11}$$

3) Feature Reconstruction Module: To explore the consistency information of views, we employ a feature reconstruction module, which is inspired by the proposition put forwards in [49]: an effective consistency representation for views can predict various perspectives of the same object independently. Specifically, we input the fused features into a series of decoders to reconstruct or predict the sample features for each view to address the proposition. This process can be described as $\widehat{X}^{(v)} = D^{(v)}(\overline{Z})$, where $\widehat{X}^{(v)} \in R^{n\times d_v}$ represents the features of samples in the v-th view predicted by the fused features and where $D^{(v)}$ denotes the decoder for reconstructing shared features in the v-th view. $D^{(v)}$ consists of several fully connected layers. This process is guided by a reconstruction loss function $\mathcal{L}_{rec}$:

$$\mathcal{L}_{rec} = \frac{1}{m}\sum_{v}^{m}\left(\sum_{i=1}^{n}\left\|X_i^{(v)} - \widehat{X_i}^{(v)}\right\|_2^2 W_{i,v}\right)/nd_v \tag{12}$$

where $\mathcal{L}_{rec}$ introduces the missing view indicator matrix W to guide the model's learning process using only the available views.

4) Incomplete Multilabel Classification Module With a Pseudolabel Strategy: To predict the label of the sample, we adopt an end-to-end multilabel classification strategy, which uses a classifier consisting of a fully connected layer and a sigmoid activation function to establish connections between the feature space and the label space: $P = \Psi(\overline{Z})$, where Ψ is the classifier and $P \in [0,1]^{n\times c}$ is the prediction probability matrix. Similarly, we also use the following weighted BCE as the objective function to explore certain available label information to guide model training:

$$\mathcal{L}_{BCE} = -\frac{1}{nc}\sum_{i=1}^{n}\sum_{j=1}^{c}\left(Y_{i,j}\log\left(P_{i,j}\right) + (1-Y_{i,j})\log\left(1-P_{i,j}\right)\right)G_{i,j} \tag{13}$$

As shown above, this approach utilizes only a portion of the labelled data for model training, as demonstrated in [50], which simply mitigates the detrimental effects of missing labels on model training by ignoring them. However, training the model on only a subset of the dataset that neglects missing labels may result in a suboptimal classification boundary. To address this issue, some researchers have suggested annotating missing labels as negatives to direct model training [26]. This approach, however, poses a significant risk: it may erroneously transform positive labels into negative labels, thereby increasing noise labels within the training set. To counteract this challenge, we

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design a novel pseudolabel strategy to identify and select credible negative labels, thereby providing more accurate guidance for model training. In general, we rewrite the weighted BCE loss to form two parts: $\mathcal{L}_{BCE} = \mathcal{L}^{l}_{BCE} + \mathcal{L}^{u}_{BCE}$, where $\mathcal{L}^{l}_{BCE}$ and $\mathcal{L}^{u}_{BCE}$ represent the loss of available labels and the loss of missing labels, respectively.

$$\mathcal{L}^{l}_{BCE} = -\frac{1}{nc}\sum_{i=1}^{n}\sum_{j=1}^{c}\left(Y_{i,j}\log\left(P_{i,j}\right) + \left(1-Y_{i,j}\right)\log\left(1-P_{i,j}\right)\right)G_{i,j} \tag{14}$$

If we wish to ignore the effect of missing labels, then $\mathcal{L}^{u}_{BCE} = 0$; if we treat the missing labels as negative labels, then $\mathcal{L}^{u}_{BCE} = \mathcal{L}^{u^-}_{BCE}$, where $\mathcal{L}^{u^-}_{BCE}$ is the loss from treating the missing labels as negative labels. Our goal is to find a trustworthy $\mathcal{L}^{u^-}_{BCE}$ to accurately guide model learning. Specifically, we assess whether the pseudolabels are trustworthy according to the loss between the existing label and the predicted label. Initially, we calculate the loss U between the complete label filled with negative pseudolabels and the prediction probability:

$$U = Y \odot \log(P) + (1-Y) \odot \log(1-P) \tag{15}$$

where $U \in R^{n\times c}$. We subsequently compute the average loss for the real labels of each sample, as delineated below:

$$\varrho_i = \frac{\sum_{j=1}^{c} G_{i,j}U_{i,j}}{\sum_{j=1}^{c} G_{i,j}} \tag{16}$$

where $\varrho \in R^{n\times 1}$ and where ϱ_i indicates the average loss of the labels existing in the i-th sample. Then, we choose the trustworthy negative label according to ϱ as follows:

$$G^{t-}_{i,j} = \begin{cases} 1 - G_{i,j} & \text{if } U_{i,j} \le \varrho_i \\ 0 & \text{otherwise} \end{cases} \tag{17}$$

where $G^{t-} \in \{0,1\}^{n\times c}$ is the indicator matrix of trustworthy negative labels. $G^{t-}_{i,j} = 1$ means that the j-th label of the i-th sample is missing, and the negative tag is selected as a trustworthy pseudolabel; otherwise, $G^{t-}_{i,j} = 0$. Therefore, we describe the trustworthy $\mathcal{L}^{u^-}_{BCE}$ as $\mathcal{L}^{t^-}_{BCE}$ and have the following formulation:

$$\mathcal{L}^{t^-}_{BCE} = \frac{sum(U \odot G^{t-})}{sum(G^{t-})} \tag{18}$$

where $sum()$ indicates the sum of all the elements in the matrix. Equation (18) selects only the trustworthy negative labels as pseudolabels and guides model learning without introducing noise information. Finally, $\mathcal{L}_{BCE} = \mathcal{L}^{l}_{BCE} + \mathcal{L}^{t^-}_{BCE}$.

Similarly, for the v-th view, we rewrite its view-specific loss $\mathcal{L}_{BCE_v}$ in (11) as follows: $\mathcal{L}_{BCE_v} = \mathcal{L}^{l}_{BCE_v} + \mathcal{L}^{t^-}_{BCE_v}$.

$$\mathcal{L}^{l}_{BCE_v} = -\frac{1}{mnc}\sum_{v=1}^{m}\sum_{i=1}^{n}\sum_{j=1}^{c}\left(Y_{i,j}\log\left(P^{(v)}_{i,j}\right) + \left(1-Y_{i,j}\right)\log\left(1-P^{(v)}_{i,j}\right)\right)H^{v}_{i,j} \tag{19}$$

We subsequently elaborate on the computation of $\mathcal{L}^{t^-}_{BCE_v}$. Initially, we determine the loss for each view by assessing the discrepancy between the predicted labels and their actual labels, as outlined below:

$$U^v = Y \odot \log(P^{(v)}) + (1-Y) \odot \log(1-P^{(v)}) \tag{20}$$

where $U^v \in R^{n\times c}$ denotes the loss matrix of the v-th view. Then, we compute the average loss of the real labels for samples within each view. Based on these calculations, the credible pseudolabel indication matrix is established as follows:

$$\varrho^v_i = \frac{\sum_{j=1}^{c} G_{i,j}U^{(v)}_{i,j}}{\sum_{j=1}^{c} G_{i,j}} \tag{21}$$

$$G^{v-}_{i,j} = \begin{cases} (1-G_{i,j})W_{i,v} & \text{if } U^{(v)}_{i,j} \le \varrho_i \\ 0 & \text{otherwise} \end{cases} \tag{22}$$

where ϱ^v_i and $G^{v-}_{i,j} \in \{0,1\}^{n\times c}$ represent the average loss of the real label of the i-th sample and the indication matrix of the selected pseudolabel in the v-th view, respectively. We also introduce the view missing indication matrix W to shield the influence of missing samples in a specific view. $\mathcal{L}^{t^-}_{BCE_v}$ is expressed as follows:

$$\mathcal{L}^{t^-}_{BCE_v} = \frac{1}{m}\sum_{v=1}^{m}\frac{sum(U^v \odot G^{v-})}{sum(G^{v-})} \tag{23}$$

However, there is a disadvantage in directly using the pseudolabel strategy to guide model learning. In the initial learning period, the model's predictions are often imprecise, and directly applying the pseudolabel strategy could inadvertently incorporate erroneous information. Therefore, we apply a clock mechanism to initiate the pseudolabel strategy at an opportune epoch.

5) Total Loss Function: The total objective function is divided into two parts, namely, classification loss and reconstruction loss. The following equation formally represents the overall objective function:

$$\mathcal{L}_{all} = \mathcal{L}_{cl} + \alpha\mathcal{L}_{rec} \tag{24}$$

where α is the penalty coefficient for reconstruction loss. $\mathcal{L}_{cl}$ is the total classification loss, calculated as $\mathcal{L}_{cl} = \mathcal{L}_{BCE} + \mathcal{L}_{BCE_v}$.

The training procedure of our model is depicted in the pseudocode of Algorithm 1. For the testing phase, the sample features are added into the model, which outputs a probability matrix P. The matrix P indicates the likelihood of each sample being associated with a particular label. The classification criteria are straightforward: if the element $P_{i,j}$ is greater than or equal to the threshold of 0.5, we deem the j-th label applicable to the i-th sample and denote this by setting the predicted label $\widehat{Y}_{i,j}$ to 1. On the other hand, if P is less than 0.5, the j-th label is not assigned, and the matrix entry $\widehat{Y}_{i,j}$ is accordingly marked as 0. This binary decision process effectively translates the probabilistic outputs into discrete label assignments.

IV. Experiments

This section meticulously details the experimental configuration, presents a systematic analysis of the outcomes, and includes an ablation study to measure the effectiveness of various

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Algorithm 1: Training Procedure of the URDF Model.

Input: The training samples' partial multiview data $\{X^{(v)}\}_{v=1}^{m}$ and the associated incomplete label matrix Y, missing-view indication matrix W, missing-label indication matrix G, batch size B, and learning rate lr; Maximum number of iterations T, hyper-parameters α, and η; Wake-up parameter ε.

for t=1 to T **do**

  for $b = 1$ **to** n/B **do**

  Constructing multiview data $\overline{X}^{(v)}$ with mask fragments in the b-th batch through the $Mask$ module;

  Extracting the high-order semantic features $\widetilde{X}^{(v)}$ in the b-th batch of views by E^v;

  Fusion of cross-view features of instances in the b-th batch by (8);

  Reconstruction of features in the b-th batch by $D^{(v)}$;

  Calculate the reconstruction loss $\mathcal{L}_{rec}$ in the b-th batch;

  if $t > \varepsilon$:

  Wake-up pseudolabel mechanism;

  end if

  Calculate the classification loss $\mathcal{L}_{cl}$ in the b-th batch;

  Calculate the total loss $\mathcal{L}_{all}$ in the b-th batch via (24);

  Update the network parameters.

  end for

end for

return model

TABLE II
THE OVERALL SITUATION OF FIVE DATASETS: n REPRESENTS THE COMPLETE COUNT OF INSTANCES PRESENT, c REPRESENTS THE SUM OF ALL LABELS, AND $\#avg$ DENOTES THE MEAN ALLOCATION OF LABELS TO EACH INSTANCE

datasets	n	c	$\#avg$
Corel5k	4999	260	3.396
Pascal07	9963	20	1.465
Espgame	20770	268	4.686
IAPRTC12	19627	291	5.719
Mirflickr	25000	38	4.716

model components. It further delves into the sensitivity of the model to different hyperparameters and confirms the model's convergence properties. These experiments ensure a reliable evaluation of the model's performance and robustness.

A. Experiment Configuration

1) Datasets: In alignment with prior studies [20], [21], [22], [37], [41], we validate our proposed model via five renowned multiview multilabel datasets: Corel5k, Pascal07, Espgame, IAPRTC12, and Mirflickr. These datasets provide a rich variety of views, including six different perspectives: GIST, HSV, HUE, SIFT, RGB, and LAB. A brief overview of these datasets is provided in Table II, with more exhaustive details to be elaborated as follows.

Corel5k [51]: The Corel5k dataset comprises a collection of 4,999 images, where each image is annotated with one to five tags selected from a comprehensive classification framework encompassing 260 unique classes.

Pascal07 [52]: The Pascal07 dataset is widely recognized and frequently utilized for tasks related to visual object detection and recognition. It includes a collection of 9,963 images that depict 20 categories of objects, allowing for the possibility of multiple objects being present within a single image.

Espgame [53]: The Espgame dataset is composed of 20,770 images collected from the interactive game environment, where each image has been annotated with a varying number of descriptive labels, ranging from 1 to 15.

IAPRTC12 [54]: The IAPRTC12 dataset is a substantial collection in the realm of image classification, comprising 19,627 picture samples that are categorized into 291 distinct classes.

Mirflickr [55]: The Mirflickr dataset is an extensive compilation of 25,000 images sourced from the Flickr platform. This collection has a total of 38 distinct tags applied across the images.

2) Constructing Incomplete Datasets: Given that the five datasets possess complete views and annotations, they are unsuitable for assessing the model's performance in scenarios with incomplete data. To replicate partial multiview and incomplete multilabel dataset conditions, as described in [20], [21], [22], [37], [41], we modify the intact dataset by introducing randomness to induce a missing data rate of $\varepsilon\%$ across each view while ensuring that each sample has a minimum of one view available, thereby creating a dataset with partial views. To simulate the scenario of incomplete multilabels, we selectively eliminate a random $\omega\%$ of the positive and negative labels from the training datasets. Missing values in the views and labels are supplemented with zeros. Each dataset is subsequently partitioned into three segments at random: 70% of the data are allocated to the training dataset, 15% of the data are reserved for the validation dataset, and the remaining 15% of the data are designated for the testing dataset.

3) Compared Methods: To assess the performance of our proposed URDF model, we chose eight popular and advanced models for comparative analysis and performed experiments across five datasets. The selected methods include *C2AE* [18], *LVSL* [32], *iMvWL* [20], *NAIML* [37], *LMVCAT* [41], *DIC-net* [22], *MTD* [23], *DD-IMvMLC-net* [21], and *RANK* [40]. Detailed descriptions of these approaches are included in Section II, and a summary of their characteristics is presented in Table III. Owing to the inability of the C2AE and LVLS models

TABLE III
HERE IS A CONCISE SUMMARY OF COMPARATIVE METHODS, HIGHLIGHTING THEIR CAPABILITIES: THE 'MULTIVIEW' AND 'PARTIALVIEW' ATTRIBUTES SIGNIFY WHETHER A METHOD IS ADAPTABLE TO MULTIVIEW OR PARTIAL MULTIVIEW SCENARIOS, WHILE THE 'INCOMPLETE LABEL' DENOTES ITS CAPACITY TO ADDRESS INCOMPLETE MULTI-LABEL CLASSIFICATION TASKS

Method	Sources	Multi-view	Partial-view	Incomplete-label
C2AE	AAAI'17	×	×	✓
LVSL	TMM'22	✓	×	×
iMvWL	IJCA'18	✓	✓	✓
NAIML	TPAMI'21	✓	✓	✓
LMVCAT	AAAI'23	✓	✓	✓
DIC-net	AAAI'23	✓	✓	✓
MTD	NeurIPS'23	✓	✓	✓
DD-IMvMLC-net	TNNLS'24	✓	✓	✓
RANK	TPAMI'25	✓	✓	✓

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TABLE IV
THE EXPERIMENTAL RESULTS FROM THE APPLICATION OF NINE DIFFERENT APPROACHES ACROSS FIVE DATASETS UNDER SETTINGS CHARACTERIZED BY A 50% ABSENCE OF VIEWS, A 50% ABSENCE OF LABELS, AND70% TRAINING SAMPLES

DATA	METRIC	C2AE	LVSL	iMvWL	NAIML	DD-IMvMLC-net	LMVCAT	DIC-net	MTD	RANK	OURS
Corel5k	AP↑	0.227±0.008	0.342±0.004	0.283±0.007	0.309±0.004	0.353±0.006	0.382±0.004	0.381±0.004	0.415±0.008	0.419±0.007	**0.446±0.005**
	1-HL↑	0.980±0.002	0.987±0.000	0.978±0.000	0.987±0.000	0.987±0.000	0.986±0.000	0.988±0.000	0.988±0.000	0.988±0.000	**0.988±0.000**
	1-RL↑	0.804±0.010	0.881±0.003	0.865±0.005	0.878±0.002	0.867±0.001	0.880±0.002	0.882±0.004	0.893±0.004	**0.911±0.003**	0.901±0.002
	AUC↑	0.806±0.010	0.884±0.003	0.868±0.003	0.881±0.002	0.870±0.001	0.883±0.002	0.884±0.004	0.896±0.004	**0.914±0.004**	0.904±0.002
	1-OE↑	0.246±0.016	0.391±0.009	0.311±0.015	0.350±0.009	0.430±0.013	0.453±0.006	0.468±0.007	0.491±0.012	0.480±0.015	**0.530±0.008**
	1-COV↑	0.596±0.016	0.718±0.006	0.702±0.008	0.725±0.005	0.690±0.003	0.727±0.006	0.727±0.011	0.749±0.009	**0.795±0.003**	0.769±0.006
Pascal07	AP↑	0.485±0.008	0.504±0.005	0.437±0.018	0.488±0.003	0.532±0.002	0.519±0.005	0.505±0.012	0.551±0.004	0.556±0.007	**0.565±0.005**
	1-HL↑	0.908±0.002	0.930±0.000	0.882±0.004	0.928±0.001	0.931±0.001	0.924±0.003	0.929±0.001	0.932±0.001	0.932±0.001	**0.934±0.001**
	1-RL↑	0.745±0.009	0.806±0.003	0.736±0.015	0.783±0.001	0.813±0.000	0.811±0.004	0.783±0.008	0.831±0.003	0.830±0.005	**0.831±0.004**
	AUC↑	0.765±0.010	0.832±0.002	0.767±0.012	0.811±0.001	0.833±0.002	0.834±0.004	0.809±0.006	**0.851±0.003**	0.851±0.005	0.848±0.004
	1-OE↑	0.438±0.008	0.419±0.008	0.362±0.023	0.421±0.006	0.452±0.006	0.422±0.006	0.427±0.015	0.459±0.007	0.470±0.012	**0.476±0.005**
	1-COV↑	0.680±0.010	0.751±0.003	0.677±0.015	0.727±0.002	0.760±0.001	0.763±0.005	0.731±0.006	0.784±0.003	0.783±0.005	**0.784±0.004**
Espgame	AP↑	0.202±0.006	0.285±0.003	0.244±0.005	0.246±0.002	0.287±0.002	0.294±0.004	0.297±0.002	0.306±0.003	0.312±0.004	**0.317±0.003**
	1-HL↑	0.971±0.002	0.983±0.000	0.972±0.000	0.983±0.000	0.983±0.000	0.983±0.000	0.983±0.000	0.983±0.000	0.983±0.000	**0.983±0.000**
	1-RL↑	0.772±0.006	0.829±0.001	0.807±0.001	0.818±0.002	0.821±0.000	0.828±0.002	0.832±0.001	0.837±0.002	**0.846±0.002**	0.843±0.001
	AUC↑	0.777±0.006	0.833±0.002	0.813±0.002	0.824±0.002	0.826±0.004	0.833±0.002	0.836±0.001	0.842±0.002	**0.850±0.002**	0.847±0.002
	1-OE↑	0.262±0.018	0.389±0.004	0.343±0.013	0.339±0.003	0.430±0.004	0.434±0.009	0.439±0.007	0.447±0.009	**0.459±0.002**	0.452±0.007
	1-COV↑	0.497±0.011	0.567±0.005	0.548±0.004	0.571±0.003	0.564±0.003	0.590±0.004	0.593±0.003	0.602±0.004	**0.625±0.005**	0.620±0.006
IAPRTC12	AP↑	0.224±0.007	0.304±0.004	0.237±0.003	0.261±0.001	0.308±0.001	0.317±0.003	0.323±0.001	0.332±0.003	0.342±0.005	**0.353±0.003**
	1-HL↑	0.965±0.002	0.981±0.000	0.969±0.000	0.981±0.000	0.980±0.000	0.980±0.000	0.981±0.000	0.981±0.000	0.981±0.005	**0.981±0.000**
	1-RL↑	0.806±0.005	0.861±0.002	0.833±0.003	0.848±0.001	0.864±0.001	0.870±0.001	0.873±0.001	0.875±0.002	**0.885±0.002**	0.882±0.002
	AUC↑	0.807±0.005	0.863±0.001	0.836±0.002	0.850±0.001	0.864±0.001	0.872±0.001	0.874±0.001	0.876±0.001	**0.886±0.002**	0.883±0.002
	1-OE↑	0.300±0.031	0.429±0.009	0.352±0.008	0.390±0.005	0.436±0.001	0.443±0.005	0.468±0.002	0.467±0.005	0.482±0.009	**0.494±0.007**
	1-COV↑	0.523±0.009	0.597±0.004	0.564±0.005	0.592±0.004	0.614±0.001	0.649±0.003	0.649±0.001	0.649±0.004	**0.677±0.005**	0.669±0.004
Mirflickr	AP↑	0.505±0.008	0.553±0.002	0.490±0.012	0.551±0.002	0.602±0.002	0.594±0.005	0.589±0.005	0.607±0.004	0.609±0.005	**0.615±0.004**
	1-HL↑	0.853±0.004	0.885±0.001	0.839±0.002	0.882±0.001	0.888±0.000	0.882±0.002	0.888±0.002	0.891±0.001	0.891±0.001	**0.892±0.000**
	1-RL↑	0.821±0.003	0.856±0.001	0.806±0.011	0.844±0.001	0.865±0.001	0.865±0.003	0.863±0.004	0.875±0.002	0.876±0.002	**0.877±0.001**
	AUC↑	0.810±0.004	0.844±0.001	0.794±0.015	0.837±0.001	0.852±0.001	0.853±0.003	0.849±0.004	0.862±0.002	**0.862±0.002**	0.860±0.002
	1-OE↑	0.505±0.020	0.607±0.004	0.511±0.022	0.585±0.003	0.684±0.001	0.643±0.008	0.637±0.007	0.655±0.004	0.657±0.007	**0.669±0.006**
	1-COV↑	0.590±0.005	0.636±0.001	0.572±0.013	0.631±0.002	0.653±0.002	0.668±0.003	0.652±0.007	0.676±0.004	0.675±0.003	**0.678±0.002**

The symbol ↑ denotes that greater values correspond to better model efficacy.

to handle PMvIMlL data directly, we employ the strategy of using the average features from existing views to impute the incomplete views and substitute missing labels with zeros.

4) Evaluation Metrics: Referring to approaches in the literature [20], [21], [22], [37], [41], we assess the performance of the URDF model through four prevalent recognized metrics within the domain of multilabel classification: average precision (AP), Hamming loss (HL), ranking loss (RL), and area under the ROC curve (AUC). To render these metrics more directly indicative of model performance, we invert the HL and RL to form 1-HL and 1-RL, ensuring that a higher numerical value across all metrics denotes superior model competence. Additionally, we use the one error (OE) and coverage (Cov) metrics, which are also commonly used in multilabel classification methods, as referenced in [56], [57]. Similarly, for the OE and Cov metrics, we also transform OE to 1-OE and Cov to 1-Cov; a higher value is indicative of better performance.

5) Implementation Details: Our model's implementation is facilitated by Python and PyTorch libraries, with a momentum value set to 0.9 for the SGD optimizer, which iteratively adjusts the network's parameters. We use a batch size of 128 and a learning rate of 1 for our training process. In the Pascal07 dataset, the pseudolabelling mechanism is set to activate at the 10th epoch, whereas for the remaining datasets, it is scheduled for the 30th epoch. The model is executed on an Ubuntu operating system capitalized on by an RTX 3080Ti GPU and an i7-12700KF CPU.

B. Experimental Results and Analysis

To prove the superiority of our method, we perform a comparative experiment with the other eight current best methods. The experimental outcomes for nine methodologies across all datasets, under conditions of 50% views and 50% labels, are presented in Table IV. We further investigate the model's performance when only a single type of missing data is addressed, with the corresponding experimental outcomes presented in Table V. Additionally, to study the influence of different masking rates on the model, we use different masking rates to test the performance of the model. Fig. 3 illustrates the performance of the URDF model on datasets subjected to varying view masking rates. We adjusted the masking rates across [0%, 25%, 50%, 75%] to assess how different levels of data occlusion impact the model's efficacy. In addition, our model is tested in the complete multi-view multilabel scenario. To ensure the reliability of our findings and to counteract the effects of randomness, each experiment was repeated a total of 10 times. The reported results reflect the average performance and the standard deviation derived from these multiple trials. According to Tables IV, V, VI, and Fig. 3, the following insights are obtained:

- As depicted in Table IV, our model obviously outperforms other algorithms in terms of the comprehensive performance of the models. Taking the average precision (AP) as an example, our model achieves a significant lead of approximately 3% over the second-ranking model, RANK, on the Corel5k dataset and approximately 2% on the IAPRTC12 dataset. These experimental results demonstrate the superior effectiveness of our model, which has delivered the best experimental outcomes to date.
- Table IV clearly shows the significant benefits of a method that considers both missing-view and missing-label scenarios in comparison to other methodologies that consider only one of the missing cases. Among the various methodologies in Table IV, C2AE and LVSL address only one of the two types of missing data, whereas the remaining approaches concurrently address both types of missing

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TABLE V
THE OUTCOMES OF THE FOUR METHODOLOGIES ACROSS FIVE DISTINCT DATASETS UNDER CONDITIONS OF FULL VIEWS WITH 50% LABEL ABSENCE AND COMPLETE LABELS WITH 50% VIEW ABSENCE

DATA	METRIC	complete views and 50% missing labels					50% missing views and complete labels				
		iMvWL	NAIML	DIC-net	RANK	OURS	iMvWL	NAIML	DIC-net	RANK	OURS
Corel5k	AP↑	0.308±0.006	0.319±0.004	0.457±0.011	0.478±0.011	**0.519±0.004**	0.289±0.009	0.317±0.003	0.419±0.006	0.442±0.009	**0.472±0.005**
	1-HL↑	0.979±0.000	0.987±0.000	0.988±0.000	0.988±0.000	**0.989±0.000**	0.979±0.000	0.987±0.000	0.988±0.000	0.988±0.000	**0.988±0.000**
	1-RL↑	0.880±0.003	0.886±0.003	0.910±0.001	0.922±0.003	**0.924±0.004**	0.870±0.003	0.885±0.003	0.892±0.001	**0.909±0.005**	0.907±0.002
	AUC↑	0.883±0.003	0.888±0.003	0.912±0.001	0.923±0.003	**0.927±0.004**	0.873±0.003	0.888±0.002	0.895±0.002	**0.912±0.005**	0.909±0.002
	1-OE↑	0.352±0.011	0.353±0.010	0.568±0.010	0.579±0.016	**0.625±0.012**	0.320±0.015	0.360±0.008	0.499±0.005	0.525±0.014	**0.561±0.015**
	1-COV↑	0.731±0.005	0.741±0.006	0.783±0.004	0.810±0.006	**0.810±0.004**	0.717±0.008	0.742±0.006	0.761±0.005	**0.790±0.011**	0.784±0.005
Pascal07	AP↑	0.470±0.006	0.493±0.003	0.562±0.007	**0.634±0.004**	0.629±0.005	0.435±0.009	0.493±0.002	0.525±0.003	0.574±0.009	**0.580±0.007**
	1-HL↑	0.888±0.001	0.929±0.001	0.934±0.000	0.937±0.001	**0.938±0.001**	0.882±0.003	0.929±0.001	0.932±0.002	0.934±0.001	**0.935±0.001**
	1-RL↑	0.763±0.006	0.790±0.002	0.837±0.005	**0.882±0.002**	0.873±0.004	0.737±0.005	0.791±0.001	0.805±0.004	**0.844±0.004**	0.841±0.005
	AUC↑	0.792±0.006	0.818±0.001	0.857±0.003	**0.897±0.003**	0.882±0.005	0.768±0.006	0.818±0.001	0.826±0.001	**0.864±0.005**	0.856±0.005
	1-OE↑	0.410±0.007	0.424±0.007	0.472±0.001	**0.544±0.008**	0.541±0.006	0.358±0.023	0.424±0.006	0.440±0.001	0.482±0.012	**0.489±0.010**
	1-COV↑	0.704±0.007	0.736±0.002	0.789±0.003	**0.841±0.004**	0.827±0.003	0.682±0.009	0.736±0.001	0.755±0.005	**0.799±0.005**	0.795±0.005
Espgame	AP↑	0.258±0.003	0.249±0.002	0.348±0.000	0.358±0.006	**0.370±0.002**	0.245±0.004	0.249±0.002	0.315±0.003	0.327±0.005	**0.339±0.003**
	1-HL↑	0.972±0.000	0.983±0.000	0.983±0.000	0.983±0.000	**0.983±0.000**	0.972±0.000	0.983±0.000	0.983±0.000	0.983±0.000	**0.983±0.000**
	1-RL↑	0.815±0.002	0.822±0.001	0.860±0.000	0.865±0.003	**0.867±0.001**	0.810±0.002	0.822±0.002	0.838±0.003	**0.853±0.001**	0.851±0.003
	AUC↑	0.820±0.002	0.827±0.001	0.865±0.000	0.870±0.002	**0.871±0.001**	0.816±0.002	0.828±0.002	0.842±0.003	**0.857±0.002**	0.854±0.003
	1-OE↑	0.389±0.011	0.343±0.004	0.509±0.005	0.522±0.008	**0.525±0.005**	0.346±0.013	0.342±0.003	0.464±0.006	0.478±0.010	**0.482±0.009**
	1-COV↑	0.560±0.002	0.579±0.003	0.645±0.000	0.661±0.006	**0.659±0.003**	0.556±0.005	0.582±0.003	0.610±0.005	**0.640±0.003**	0.636±0.005
IAPRTC12	AP↑	0.250±0.002	0.265±0.001	0.368±0.004	0.395±0.007	**0.410±0.007**	0.234±0.005	0.265±0.001	0.347±0.002	0.355±0.004	**0.370±0.003**
	1-HL↑	0.969±0.000	0.981±0.000	0.981±0.000	0.981±0.000	**0.982±0.000**	0.968±0.000	0.981±0.000	0.981±0.000	0.981±0.000	**0.981±0.000**
	1-RL↑	0.841±0.002	0.851±0.001	0.896±0.001	**0.905±0.002**	0.904±0.002	0.832±0.003	0.851±0.001	0.881±0.001	0.891±0.002	**0.893±0.002**
	AUC↑	0.842±0.002	0.853±0.001	0.896±0.002	0.905±0.002	**0.905±0.002**	0.835±0.002	0.853±0.001	0.882±0.002	0.891±0.002	**0.893±0.002**
	1-OE↑	0.378±0.010	0.394±0.004	0.525±0.001	0.556±0.012	**0.564±0.009**	0.351±0.009	0.393±0.005	0.491±0.014	0.494±0.008	**0.508±0.004**
	1-COV↑	0.578±0.005	0.600±0.003	0.691±0.005	**0.720±0.004**	0.714±0.008	0.567±0.004	0.602±0.003	0.670±0.005	0.695±0.006	**0.702±0.006**
Mirflickr	AP↑	0.502±0.020	0.554±0.001	0.648±0.001	0.666±0.004	**0.673±0.003**	0.490±0.010	0.553±0.002	0.588±0.003	0.613±0.006	**0.622±0.003**
	1-HL↑	0.843±0.006	0.882±0.001	0.900±0.000	0.903±0.001	**0.904±0.001**	0.863±0.002	0.882±0.001	0.890±0.000	0.893±0.001	**0.894±0.001**
	1-RL↑	0.812±0.020	0.846±0.001	0.892±0.002	**0.902±0.002**	0.901±0.002	0.806±0.007	0.846±0.001	0.866±0.004	0.879±0.002	**0.880±0.001**
	AUC↑	0.795±0.030	0.839±0.001	0.875±0.003	0.885±0.002	**0.883±0.002**	0.797±0.012	0.838±0.001	0.852±0.001	0.865±0.002	**0.864±0.002**
	1-OE↑	0.523±0.018	0.588±0.003	0.719±0.001	0.730±0.006	**0.744±0.004**	0.513±0.019	0.588±0.003	0.638±0.003	0.664±0.008	**0.681±0.004**
	1-COV↑	0.577±0.029	0.634±0.002	0.696±0.003	**0.715±0.003**	0.714±0.006	0.585±0.011	0.634±0.002	0.661±0.003	0.683±0.004	**0.685±0.003**

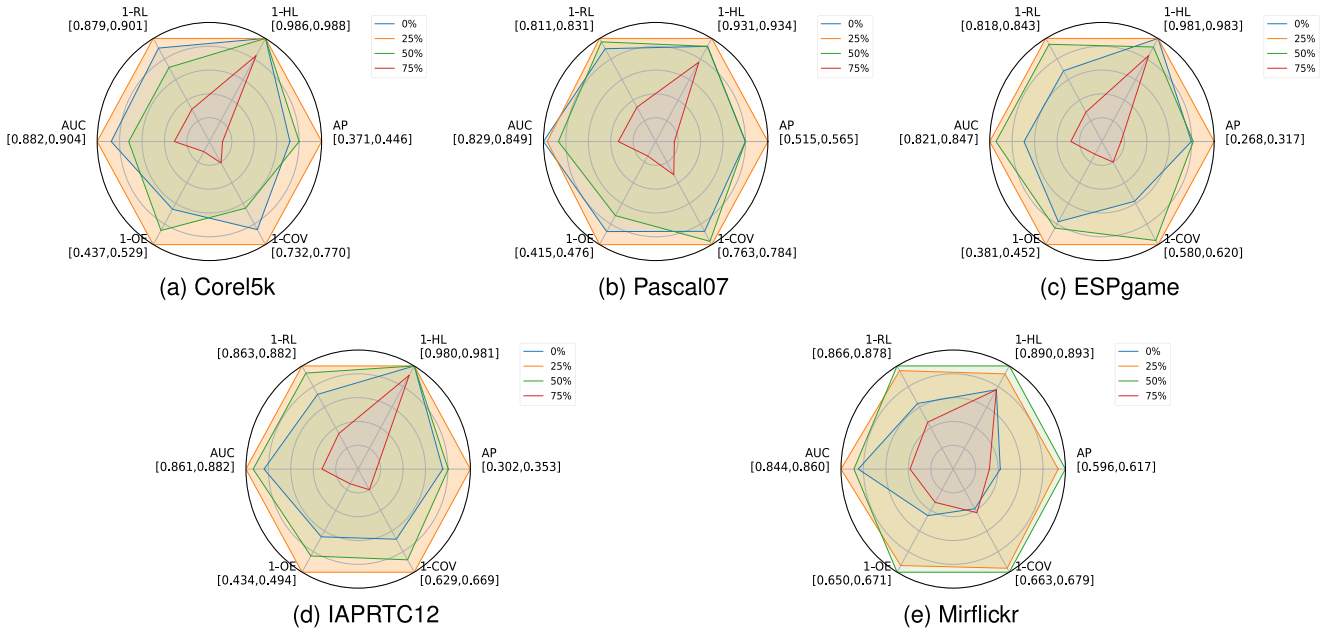


Fig. 3. The experimental results of the URDF model across five datasets with varying masking rates are depicted on a radar map. The centre indicates the poorest outcomes, whereas the vertices represent superior performance across six metrics.

data. This finding indicates that simultaneously considering multiple partial views and incomplete multilabels is not only critical for practical application but also beneficial for model performance.

- As delineated in Table V, our approach is also capable of handling the absence of a single type of missing data, and the comprehensive performance of our model is the highest among several methods under the conditions of this scenario. Additionally, the performances of all the methodologies are more adversely affected in the scenario where the view is absent than in the scenario where the label is missing. For example, on the Corel5k dataset, the AP metric of our model is 47.2% under conditions where there is a missing view alongside a complete view

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TABLE VI
THE EXPERIMENTAL RESULTS FROM THE APPLICATION OF NINE DIFFERENT APPROACHES ACROSS FIVE DATASETS UNDER COMPLETE VIEWS AND LABELS, WITH 70% TRAINING SAMPLES

DATA	METRIC	C2AE	LVSL	iMvWL	NAIML	DD-IMvMLC-net	LMVCAT	DIC-net	MTD	RANK	OURS
Corel5k	AP↑	0.353±0.033	0.482±0.003	0.313±0.005	0.327±0.004	0.486±0.015	0.521±0.005	0.509±0.002	0.523±0.003	0.524±0.008	**0.554±0.004**
	1-HL↑	0.969±0.002	0.988±0.000	0.979±0.000	0.987±0.000	0.988±0.000	0.988±0.000	0.989±0.000	0.989±0.000	0.989±0.000	**0.989±0.000**
	1-RL↑	0.870±0.020	0.927±0.001	0.884±0.002	0.890±0.002	0.918±0.008	0.928±0.001	0.929±0.001	0.931±0.001	0.932±0.003	**0.933±0.002**
	AUC↑	0.873±0.020	0.930±0.001	0.887±0.002	0.893±0.002	0.921±0.007	0.930±0.002	0.931±0.001	0.934±0.001	0.934±0.003	**0.935±0.002**
	1-OE↑	0.434±0.042	0.554±0.003	0.346±0.010	0.371±0.009	0.593±0.017	0.618±0.008	0.615±0.007	0.629±0.007	0.627±0.013	**0.655±0.006**
	1-COV↑	0.726±0.034	0.819±0.003	0.739±0.005	0.751±0.005	0.797±0.018	0.827±0.003	0.822±0.001	0.830±0.003	**0.834±0.006**	0.831±0.005
Pascal07	AP↑	0.584±0.007	0.588±0.006	0.468±0.004	0.496±0.003	0.648±0.006	0.629±0.007	0.608±0.003	0.648±0.006	**0.656±0.007**	0.654±0.006
	1-HL↑	0.921±0.002	0.934±0.001	0.888±0.001	0.929±0.001	0.940±0.001	0.927±0.005	0.937±0.001	0.940±0.001	0.940±0.001	**0.941±0.001**
	1-RL↑	0.835±0.006	0.871±0.003	0.763±0.007	0.795±0.001	0.885±0.004	0.878±0.004	0.859±0.003	0.886±0.003	**0.891±0.004**	0.884±0.003
	AUC↑	0.851±0.006	0.891±0.002	0.793±0.005	0.822±0.001	0.896±0.003	0.892±0.003	0.876±0.002	0.897±0.003	**0.902±0.005**	0.892±0.002
	1-OE↑	0.510±0.009	0.483±0.008	0.412±0.005	0.426±0.006	0.564±0.008	0.534±0.011	0.526±0.005	0.562±0.008	**0.574±0.013**	0.569±0.010
	1-COV↑	0.783±0.007	0.825±0.003	0.704±0.007	0.741±0.002	0.842±0.003	0.836±0.003	0.815±0.004	0.844±0.003	**0.848±0.005**	0.840±0.002
Espgame	AP↑	0.269±0.003	0.363±0.002	0.260±0.004	0.251±0.002	0.383±0.004	0.385±0.003	0.391±0.001	0.397±0.002	0.388±0.003	**0.404±0.005**
	1-HL↑	0.958±0.002	0.983±0.000	0.972±0.000	0.983±0.000	0.983±0.000	0.982±0.000	0.984±0.000	0.984±0.000	0.984±0.000	**0.984±0.000**
	1-RL↑	0.832±0.002	0.836±0.001	0.817±0.001	0.825±0.001	0.869±0.004	0.876±0.002	0.871±0.003	**0.880±0.001**	0.877±0.001	0.877±0.002
	AUC↑	0.837±0.002	0.838±0.001	0.822±0.001	0.830±0.002	0.872±0.004	0.880±0.002	0.874±0.004	**0.883±0.001**	0.881±0.000	0.880±0.002
	1-OE↑	0.354±0.006	0.505±0.004	0.390±0.011	0.345±0.003	0.534±0.011	0.540±0.005	0.549±0.002	**0.554±0.005**	0.545±0.006	0.546±0.006
	1-COV↑	0.614±0.005	0.577±0.003	0.566±0.003	0.587±0.003	0.666±0.008	0.687±0.004	0.674±0.006	**0.693±0.002**	0.688±0.004	0.684±0.005
IAPRTC12	AP↑	0.316±0.004	0.383±0.003	0.250±0.004	0.267±0.001	0.414±0.004	0.436±0.004	0.418±0.006	0.443±0.003	0.428±0.005	**0.451±0.004**
	1-HL↑	0.953±0.002	0.982±0.000	0.969±0.000	0.980±0.000	0.982±0.000	0.981±0.000	0.982±0.000	0.982±0.000	0.982±0.000	**0.982±0.000**
	1-RL↑	0.868±0.001	0.872±0.002	0.842±0.002	0.854±0.001	0.905±0.003	0.918±0.001	0.912±0.003	**0.918±0.002**	0.914±0.002	0.917±0.003
	AUC↑	0.869±0.002	0.873±0.001	0.843±0.002	0.855±0.001	0.907±0.002	0.918±0.001	0.911±0.003	**0.918±0.002**	0.915±0.003	0.917±0.003
	1-OE↑	0.441±0.010	0.505±0.006	0.380±0.010	0.396±0.004	0.561±0.005	0.581±0.007	0.581±0.001	0.601±0.005	0.581±0.008	**0.604±0.005**
	1-COV↑	0.652±0.002	0.620±0.003	0.581±0.005	0.608±0.003	0.718±0.006	**0.754±0.002**	0.731±0.009	0.751±0.005	0.745±0.005	0.748±0.008
Mirflickr	AP↑	0.593±0.004	0.638±0.002	0.493±0.022	0.555±0.002	0.679±0.005	0.684±0.006	0.659±0.004	0.681±0.004	0.680±0.005	**0.687±0.002**
	1-HL↑	0.869±0.002	0.895±0.001	0.840±0.004	0.882±0.001	0.905±0.001	0.897±0.005	0.902±0.001	0.906±0.001	0.905±0.001	**0.906±0.001**
	1-RL↑	0.854±0.003	0.890±0.001	0.806±0.015	0.847±0.001	0.904±0.003	0.905±0.003	0.895±0.001	0.907±0.002	0.907±0.002	**0.907±0.002**
	AUC↑	0.845±0.003	0.875±0.001	0.789±0.002	0.839±0.001	0.886±0.004	0.889±0.002	0.876±0.002	0.888±0.002	**0.890±0.002**	0.887±0.002
	1-OE↑	0.641±0.020	0.714±0.004	0.518±0.040	0.590±0.003	0.746±0.009	0.747±0.007	0.728±0.008	0.749±0.005	0.747±0.007	**0.757±0.003**
	1-COV↑	0.642±0.005	0.691±0.002	0.579±0.021	0.636±0.002	0.718±0.003	0.727±0.003	0.702±0.003	**0.724±0.002**	0.722±0.004	0.723±0.003

The symbol ↑ denotes that greater values correspond to better model efficacy.

scenario. In contrast, the AP metric of our model reaches 51.9% when faced with a scenario involving a missing label in conjunction with a complete view. This finding indicates that the absence of a view has a more pronounced effect on the model than does the absence of a label. This phenomenon may be attributed to the fact that a missing view directly influences the feature extraction and data representation processes, whereas a missing tag, although leading to information loss, predominantly affects only a subset of the categorical information pertaining to the sample and does not compromise the feature representation of the sample.

- The outcomes presented in Table VI delineate the performance across various methods within a complete multiview multilabel scenario. Compared with the other models, the overall performance of our model is the best in terms of six metrics, which means that despite our model being designed for partial multiview incomplete multilabel learning, our model also demonstrates a robust ability to effectively address complete multiview multilabel learning tasks.
- As illustrated in Fig. 3 for all datasets, the model's performance peaks at a 25% masking rate and is slightly lower, yet still strong, at a 0% masking rate. As the masking rate increases to 50% and 75%, the model's performance sequentially decreases. This trend suggests that a modest amount of masking on the original data, which removes some redundant information, can enhance the quality of the features extracted by the network. However, excessive masking removes not only unnecessary data but also valuable information, adversely affecting the model's ability to learn. Hence, it is crucial to select an appropriate masking rate to strike a balance between removing noise and preserving valuable features.

TABLE VII
THE RESULTS OF THE ABLATION STUDY OF THE URDF MODEL ON THE COREL5K AND PASCAL07 DATASETS

Backbone	$Mask$	RDF	pso_lable	Corel5k AP	Corel5k AUC	Pascal07 AP	Pascal07 AUC
✓				0.405	0.892	0.546	0.846
✓	✓			0.428	0.896	0.548	0.846
✓		✓		0.417	0.894	0.548	0.845
✓			✓	0.429	0.901	0.548	0.847
✓		✓	✓	0.421	0.900	0.549	0.848
✓	✓	✓		0.443	0.898	0.561	0.846
✓	✓		✓	0.429	0.901	0.550	0.847
✓	✓	✓	✓	0.446	0.904	0.565	0.848
pseudolabel filling rate				0.895		0.718	

C. Ablation Study

To evaluate the efficacy of the distinct modules within our model, we performed ablation experiments on the Corel5k and Pascal07 datasets under the following specific conditions: — 50% missing views, 50% missing labels, and a training set comprising 70% of the total samples. We identified four core components of our model: the backbone network and the random mask module $Mask$. The RDF (reliable dynamic fusion) module aims to increase the robustness and accuracy of the model by dynamically adjusting the fusion weights of features from different views. We added these modules to the backbone and meticulously recorded the consequent effects on the AP and AUC metrics after the addition of each module, as detailed in Table VII. Notably, in the ablation study, to assess the specific

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contribution of the RDF module to the model's performance, we employed an average weighting strategy as a substitute for the uncertainty-driven reliable sample-level dynamic fusion strategy. Specifically, we ceased dynamically adjusting the feature fusion weights based on the uncertainty and credibility of the samples, opting instead for a simple average weighting of features across all views. Additionally, we refrained from using the classification loss of specific views to ensure the fairness and comparability of the ablation study. The data in Table VII clearly demonstrate that the addition of each module significantly improves the model's performance, underscoring the positive contribution of each component to the overall effectiveness of our model.

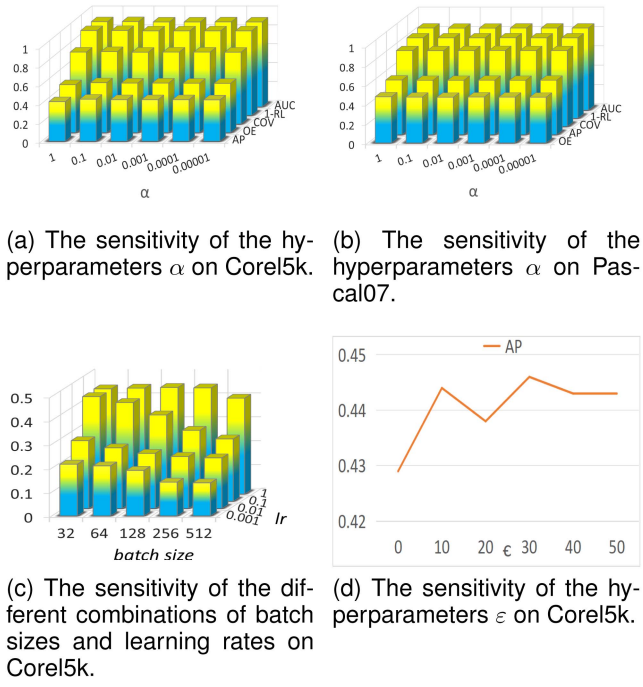
To further validate the effectiveness of the pseudolabel learning strategy, we conducted additional ablation experiments to analyse the ratio of learned pseudolabels to missing labels, as presented in Table VII. The pseudolabel filling rate indicates the average proportion of missing labels that were filled by the pseudolabels learned during the training phase by the model. As shown in Table VII, the pseudolabel filling rates are 89.5% on the Corel5k dataset and 71.8% on the Pascal07 dataset. Additionally, through ablation experiments, the pseudolabel learning strategy yielded a more significant improvement on the Corel5k dataset than on the Pascal07 dataset. This disparity may be attributed to the model's ability to learn a greater number of reliable pseudolabels on the Corel5k dataset than on the Pascal dataset, thereby making the pseudolabel learning strategy more effective on Corel5k. Furthermore, since we selected trustworthy negative labels as supervisory information for our model and negative labels generally provide less discriminative information than positive labels do, this might have resulted in relatively limited improvement from the pseudolabel learning strategy. Future work will focus on addressing this issue to further enhance the benefits of pseudolabel learning.

D. Sensitivity Studies of Hyperparameters

The objective of performing a sensitivity study on the model's hyperparameters is to determine the impact of fluctuations in these parameters on the model's efficacy. This analysis is instrumental in identifying the hyperparameters that are most crucial and necessitate fine-tuning to increase the model's accuracy, efficiency, and generalizability to new data. Furthermore, sensitivity studies can demonstrate the model's resilience across diverse hyperparameter configurations, thereby aiding in the formulation of informed decisions regarding hyperparameter selection.

We performed a sensitivity analysis of the critical hyperparameters within our model: α. Our focus was on scrutinizing the impact of modifying the values of the penalty coefficient within the loss function. We tested a range of values for α, specifically [1,0.1, 0.01, 0.001, 0.0001, 0.00001]. The resulting effects on the AP metric, OE metric, COV metric, 1-RL metric, and AUC metric for both the Corel5k and Pascal07 datasets across different values of α are depicted in Fig. 4(a)–(b).

The figure clearly shows that the optimal value for maximizing performance on the Corel5k dataset is $\alpha = 0.1$, and the optimal setting for the Pascal07 dataset is $\alpha = 1$. Furthermore, we find that the model's performance is not significantly affected by changes in the values of α, highlighting the model's inherent robustness.



(a) The sensitivity of the hyperparameters α on Corel5k.

(b) The sensitivity of the hyperparameters α on Pascal07.

(c) The sensitivity of the different combinations of batch sizes and learning rates on Corel5k.

(d) The sensitivity of the hyperparameters ε on Corel5k.

Fig. 4. A sensitivity analysis of the hyperparameters is performed on the URDF model across the Corel5k and Pascal07 datasets.

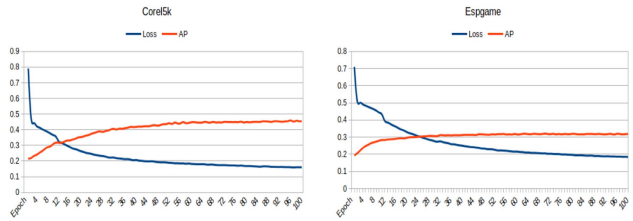
In addition, we conducted an experimental analysis to investigate the sensitivity of our model to different batch sizes and learning rates and to investigate the effects of different epochs of pseudolabel wakeup strategies on model performance. The results of these analyses are depicted in Fig. 4(c) and (d). Fig. 4(c) clearly shows that as the learning rate decreases, the performance of our model sharply decreases; this indicates the high sensitivity of our model to the learning rate, likely because a smaller learning rate can easily lead the model to converge to a local optimum rather than the global optimum. Additionally, the figure demonstrates that our model is less sensitive to variations in batch size, suggesting a greater robustness to this parameter.

Fig. 4(d) illustrates the impact of the pseudolabel activation strategy on model performance. The initial use of pseudolabels results in relatively lower model performance. However, when the pseudolabel activation strategy is employed, there is a noticeable improvement in model performance. Furthermore, the figure indicates that different rounds of pseudolabel activation have little impact on the model, suggesting that the model has a high level of robustness to this parameter.

E. Convergence Study

To evaluate the convergence characteristics of the URDF model, we performed experiments utilizing the Corel5k and Espgame datasets. Fig. 5 illustrates the changes in the objective

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(a) Convergence of the URDF model on Corel5k. (b) Convergence of the URDF model on Espgame.

Fig. 5. An investigation into the convergence of our model on both the Corel5k and Espgame datasets.

loss metric and the average precision (AP) metric for the URDF model across training epochs. As shown in Fig. 5, there is a clear and consistent trend: the objective loss decreases monotonically while the AP increases, both of which continue until the model reaches stability. These experimental outcomes substantiate the strong convergence performance of the URDF model.

V. Conclusion

In this paper, we propose a novel uncertainty-driven reliable dynamic fusion model, named URDF, to address the PMvIMlL problem. The main module of URDF model is an innovative uncertainty-driven reliable sample-level dynamic fusion module. In this fusion module, we first propose a method to assess the uncertainty of sample categories, which then links the reliability of sample features to the uncertainty of the samples. This method is intuitive: the higher the uncertainty of the sample categories is, the lower the reliability of the sample features. Furthermore, we propose an innovative pseudolabelling strategy aimed at identifying reliable negative labels to serve as pseudolabels, thereby augmenting the model's supervisory signals. Additionally, we propose a masking approach designed to bolster the encoder's learning capabilities. Finally, the experimental results on five benchmark datasets validate the superior performance of our model, which achieves state-of-the-art results.

References

[1] J. Zhao, X. Xie, X. Xu, and S. Sun, "Multi-view learning overview: Recent progress and new challenges," *Inf. Fusion*, vol. 38, pp. 43–54, 2017.
[2] X. Liu et al., "Efficient and effective regularized incomplete multi-view clustering," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 8, pp. 2634–2646, Aug. 2021.
[3] L. Li and H. He, "Bipartite graph based multi-view clustering," *IEEE Trans. Knowl. Data Eng.*, vol. 34, no. 7, pp. 3111–3125, Jul. 2022.
[4] Y. Wang, Y. Zhao, Z. Wang, C. Zhang, and X. Wang, "Robust multi-graph multi-label learning with dual-granularity labeling," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 10, pp. 6509–6524, Oct. 2024.
[5] W. Zhuge, T. Luo, R. Fan, H. Tao, C. Hou, and D. Yi, "Absent multi-view semisupervised classification," *IEEE Trans. Cybern.*, vol. 54, no. 3, pp. 1708–1721, Mar. 2024.
[6] A. M. Elkahky, Y. Song, and X. He, "A multi-view deep learning approach for cross domain user modeling in recommendation systems," in *Proc. 24th Int. Conf. World Wide Web*, 2015, pp. 278–288.
[7] H. Yang et al., "RNA-binding protein recognition based on multi-view deep feature and multi-label learning," *Brief. Bioinf.*, vol. 22, no. 3, 2021, Art. no. bbaa174.
[8] S. Yang, C. Lian, Z. Zeng, B. Xu, J. Zang, and Z. Zhang, "A multi-view multi-scale neural network for multi-label ECG classification," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 7, no. 3, pp. 648–660, Jun. 2023.
[9] M. Liu, Y. Luo, D. Tao, C. Xu, and Y. Wen, "Low-rank multi-view learning in matrix completion for multi-label image classification," in *Proc. AAAI Conf. Artif. Intell.*, 2015, pp. 2778–2784.
[10] C. Zhang, Z. Yu, Q. Hu, P. Zhu, X. Liu, and X. Wang, "Latent semantic aware multi-view multi-label classification," in *Proc. AAAI Conf. Artif. Intell.*, 2018, pp. 4414–4421.
[11] S. Sun and D. Zong, "LCBM: A multi-view probabilistic model for multi-label classification," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 8, pp. 2682–2696, Aug. 2021.
[12] J. Wen et al., "A survey on incomplete multiview clustering," *IEEE Trans. Syst., Man, Cybern. Syst.*, vol. 53, no. 2, pp. 1136–1149, Feb. 2023.
[13] W. Zhuge, H. Tao, T. Luo, L.-L. Zeng, C. Hou, and D. Yi, "Joint representation learning and clustering: A framework for grouping partial multiview data," *IEEE Trans. Knowl. Data Eng.*, vol. 34, no. 8, pp. 3826–3840, Aug. 2022.
[14] Q. Li, T. Luo, M. Jiang, J. Liao, and Z. Jiang, "Deep incomplete multi-view network semi-supervised multi-label learning with unbiased loss," in *Proc. 32nd ACM Int. Conf. Multimedia*, 2024, pp. 9048–9056.
[15] X. Fang, Y. Hu, P. Zhou, and D. Wu, "ANIMC: A soft approach for autoweighted noisy and incomplete multiview clustering," *IEEE Trans. Artif. Intell.*, vol. 3, no. 2, pp. 192–206, Apr. 2022.
[16] C. Xu, D. Tao, and C. Xu, "Multi-view learning with incomplete views," *IEEE Trans. Image Process.*, vol. 24, no. 12, pp. 5812–5825, Dec. 2015.
[17] C. Zhang et al., "CPM-Nets: Cross partial multi-view networks," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2019, pp. 577–587.
[18] C.-K. Yeh, W.-C. Wu, W.-J. Ko, and Y.-C. F. Wang, "Learning deep latent space for multi-label classification," in *Proc. AAAI Conf. Artif. Intell.*, 2017, pp. 2838–2844.
[19] Z. Ma and S. Chen, "Expand globally, shrink locally: Discriminant multi-label learning with missing labels," *Pattern Recognit.*, vol. 111, 2021, Art. no. 107675.
[20] Q. Tan, G. Yu, C. Domeniconi, J. Wang, and Z. Zhang, "Incomplete multi-view weak-label learning," in *Proc. Int. Joint Conf. Artif. Intell.*, 2018, pp. 2703–2709.
[21] J. Wen et al., "Deep double incomplete multi-view multi-label learning with incomplete labels and missing views," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 8, pp. 11396–11408, Aug. 2024.
[22] C. Liu, J. Wen, X. Luo, C. Huang, Z. Wu, and Y. Xu, "DICNet: Deep instance-level contrastive network for double incomplete multi-view multi-label classification," in *Proc. AAAI Conf. Artif. Intell.*, 2023, pp. 8807–8815.
[23] C. Liu et al., "Masked two-channel decoupling framework for incomplete multi-view weak multi-label learning," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2024, pp. 32387–32400.
[24] T. Durand, N. Mehrasa, and G. Mori, "Learning a deep ConvNet for multi-label classification with partial labels," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 647–657.
[25] D. Huynh and E. Elhamifar, "Interactive multi-label CNN learning with partial labels," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 9423–9432.
[26] K. Kundu and J. Tighe, "Exploiting weakly supervised visual patterns to learn from partial annotations," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2020, pp. 561–572.
[27] X. Fang, A. Easwaran, B. Genest, and P. N. Suganthan, "Your data is not perfect: Towards cross-domain out-of-distribution detection in class-imbalanced data," *Expert Syst. Appl.*, vol. 267, 2025, Art. no. 126031.
[28] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 16000–16009.
[29] Z. Tong, Y. Song, J. Wang, and L. Wang, "VideoMAE: Masked autoencoders are data-efficient learners for self-supervised video pre-training," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2022, pp. 10078–10093.
[30] Z. Hou et al., "GraphMAE: Self-supervised masked graph autoencoders," in *Proc. 28th ACM SIGKDD Conf. Knowl. Discov. Data Mining*, 2022, pp. 594–604.
[31] D. Zhao, Q. Gao, Y. Lu, D. Sun, and Y. Cheng, "Consistency and diversity neural network multi-view multi-label learning," *Knowl.-Based Syst.*, vol. 218, 2021, Art. no. 106841.
[32] D. Zhao, Q. Gao, Y. Lu, and D. Sun, "Non-aligned multi-view multi-label classification via learning view-specific labels," *IEEE Trans. Multimedia*, vol. 25, pp. 7235–7247, 2023.

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[33] Z. Xue, J. Du, D. Du, W. Ren, and S. Lyu, "Deep correlated predictive subspace learning for incomplete multi-view semi-supervised classification," in *Proc. Int. Joint Conf. Artif. Intell.*, 2019, pp. 4026–4032.
[34] C. Zhang, Y. Cui, Z. Han, J. T. Zhou, H. Fu, and Q. Hu, "Deep partial multi-view learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 5, pp. 2402–2415, May 2022.
[35] Y. Zhu, J. T. Kwok, and Z.-H. Zhou, "Multi-label learning with global and local label correlation," *IEEE Trans. Knowl. Data Eng.*, vol. 30, no. 6, pp. 1081–1094, Jun. 2018.
[36] C. Zhu, D. Miao, Z. Wang, R. Zhou, L. Wei, and X. Zhang, "Global and local multi-view multi-label learning," *Neurocomputing*, vol. 371, pp. 67–77, 2020.
[37] X. Li and S. Chen, "A concise yet effective model for non-aligned incomplete multi-view and missing multi-label learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 10, pp. 5918–5932, Oct. 2022.
[38] C. Liu et al., "Attention-induced embedding imputation for incomplete multi-view partial multi-label classification," in *Proc. AAAI Conf. Artif. Intell.*, 2024, pp. 13864–13872.
[39] J. Long, Q. Zhang, X. Lu, J. Wen, L. Zhao, and W. Xie, "Multi-scale locality preserving projection for partial multi-view incomplete multi-label learning," *Neural Netw.*, vol. 180, 2024, Art. no. 106748.
[40] C. Liu, J. Wen, Y. Xu, B. Zhang, L. Nie, and M. Zhang, "Reliable representation learning for incomplete multi-view missing multi-label classification," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 6, pp. 4940–4956, Jun. 2025.
[41] C. Liu, J. Wen, X. Luo, and Y. Xu, "Incomplete multi-view multi-label learning via label-guided masked view-and category-aware transformers," in *Proc. AAAI Conf. Artif. Intell.*, 2023, pp. 8816–8824.
[42] S. Ou, Z. Xue, Y. Li, M. Liang, Y. Cai, and J. Wu, "View-category interactive sharing transformer for incomplete multi-view multi-label learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 27467–27476.
[43] W. Xie et al., "Uncertainty-aware pseudo-labeling and dual graph driven network for incomplete multi-view multi-label classification," in *Proc. 32nd ACM Int. Conf. Multimedia*, 2024, pp. 6656–6665.
[44] W. Yan et al., "GCFAGG: Global and cross-view feature aggregation for multi-view clustering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 19863–19872.
[45] X. Wu et al., "Multi-view multi-label learning with view-specific information extraction," in *Proc. Int. Joint Conf. Artif. Intell.*, 2019, pp. 3884–3890.
[46] Y. Chen, Z. Wu, Z. Chen, M. Dong, and S. Wang, "Joint learning of feature and topology for multi-view graph convolutional network," *Neural Netw.*, vol. 168, pp. 161–170, 2023.
[47] J. Wen et al., "DIMC-net: Deep incomplete multi-view clustering network," in *Proc. 28th ACM Int. Conf. Multimedia*, 2020, pp. 3753–3761.
[48] Z. Chen, L. Fu, J. Yao, W. Guo, C. Plant, and S. Wang, "Learnable graph convolutional network and feature fusion for multi-view learning," *Inf. Fusion*, vol. 95, pp. 109–119, 2023.
[49] G. Ke, B. Wang, X. Wang, and S. He, "Rethinking multi-view representation learning via distilled disentangling," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 26774–26783.
[50] E. Ben-Baruch et al., "Multi-label classification with partial annotations using class-aware selective loss," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 4764–4772.
[51] P. Duygulu, K. Barnard, J. F. de Freitas, and D. A. Forsyth, "Object recognition as machine translation: Learning a lexicon for a fixed image vocabulary," in *Proc. 7th Eur. Conf. Comput. Vis.*, Springer, 2002, pp. 97–112.
[52] M. Everingham, L. Van Gool, C. Williams, J. Winn, and A. Zisserman, "The pascal visual object classes challenge 2007 (VOC2007) results http://www.pascal-network.org/challenges," 2007. [Online]. Available: VOC/voc2007/workshop/index.html
[53] L. Von Ahn and L. Dabbish, "Labeling images with a computer game," in *Proc. SIGCHI Conf. Hum. Factors Comput. Syst.*, 2004, pp. 319–326.
[54] H. Müller, M. Grubinger, T. Deselaers, and P. Clough, "The IAPR benchmark: A new evaluation resource for visual information systems," in *Proc. Int. Conf. Lang. Resour. Eval.*, 2006, pp. 1–11.
[55] M. J. Huiskes and M. S. Lew, "The MIR Flickr retrieval evaluation," in *Proc. 1st ACM Int. Conf. Multimedia Inf. Retrieval*, 2008, pp. 39–43.
[56] M.-L. Zhang and Z.-H. Zhou, "A review on multi-label learning algorithms," *IEEE Trans. Knowl. Data Eng.*, vol. 26, no. 8, pp. 1819–1837, Aug. 2014.
[57] Z.-M. Chen, X.-S. Wei, P. Wang, and Y. Guo, "Multi-label image recognition with graph convolutional networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 5177–5186.



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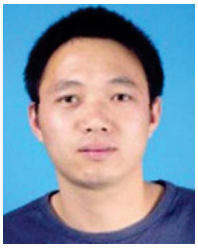


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